

SYED MUHAMMAD USMAN ZAFAR

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EDUCATION

2022-2026	NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES (FAST-NU) Bachelor of Science in Computer Science, June 2026	Lahore
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EXPERIENCE

Nov 2025 - Present	ARCOMPSOL <i>Backend Engineer</i> - Built a data pipeline for Sertis achieving >90% accuracy in automated habitational file processing. - Architected a dynamic metadata and RAG-based mapping system improving auto-mapping accuracy by 30% while reducing processing failures by 60% through Pandas optimization. - Implemented SOC 2-aligned password and authentication controls and improved infrastructure by migrating from S3 to local static storage to reduce development latency. Technologies: Python, Django, Django REST Framework, PostgreSQL, Pandas, RAG	Lahore
Jun-Aug 2025	DROPSHOT <i>Software Engineering Intern</i> - Built React/TypeScript dashboard pages from Figma designs for padel analytics, and reduced average page load times by ~40% through straightforward frontend optimizations Technologies: React, TypeScript, JavaScript, Figma	Lahore
Jun-Aug 2025	FAST School of Computing (FAST-NUCES) <i>AI Systems Intern & Team Lead</i> - Led the development of PaperPilot, an AI research platform utilizing multi-agent workflows for LLM-based extraction and automated citation for 50+ beta users. Technologies: Python, LLMs, Multi-Agent Systems, Streamlit, NetworkX, Pandas	Lahore

PROJECTS

GYNAI

AI-Driven Maternal Healthcare Decision Support System -Final Year Project (2025-Present)

- Building an AI-based clinical decision support system utilizing knowledge-graph insights to predict delivery modes and developing comprehensive doctor/patient risk-assessment dashboards
- Technologies:** React Native, TypeScript, Python, FastAPI, PostgreSQL, Neo4j

PATIENT INFLUX ANALYSIS DASHBOARD

Mansoorah Hospital (2025)

- Developed a Python-based analytics dashboard for **31** departments, reducing manual reporting time by **50%**, enabling decision-ready insights (internal, confidential deployment).
- Technologies:** Python, Streamlit, Pandas, Plotly, NumPy

ZAKAT FLOW

Automated Zakat Computation System (2024)

- Developed a Zakat automation tool that streamlined financial calculations and reduced manual effort by 90%, making it easier for community members to manage Zakat during Ramadan.
- Technologies:** Python, Streamlit, Plotly, Pandas, NumPy, forex-python

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java, Kotlin, SQL, C++

Backend & APIs: Django, Django REST Framework, FastAPI, REST APIs

Data & AI: Pandas, NumPy, Machine Learning, Knowledge Graphs, RAG, NetworkX

Databases: PostgreSQL, MongoDB, Neo4j, Firebase

Software & Tools: Git/GitHub, Linux/Unix, Postman, Android Studio